

Event Management APIs

Complete documentation for event creation, management, and discovery endpoints.

Overview

The Events API provides comprehensive functionality for:

- **Event Discovery:** Browse and search events publicly
- **Event Creation:** Vendors can create and manage events
- **Event Administration:** Admin approval and featured event management
- **Analytics:** Event performance tracking for vendors

Public Event Discovery

GET /api/events

Browse all approved events with filtering and search capabilities.

Authentication: Not required

Query Parameters:

- `page` (number): Page number (default: 1)
- `limit` (number): Items per page (default: 12, max: 50)
- `category` (string): Filter by category (e.g., "Music", "Family & Kids")
- `type` (string): Event type ("Event", "Course", "Venue")
- `venueType` (string): Venue type ("Indoor", "Outdoor")
- `city` (string): Filter by city (case-insensitive)
- `minPrice` (number): Minimum price filter
- `maxPrice` (number): Maximum price filter
- `currency` (string): Currency filter ("AED", "USD", "EGP", "CAD")
- `ageMin` (number): Minimum age range filter
- `ageMax` (number): Maximum age range filter
- `featured` (string): Filter featured events ("true" for featured only)
- `search` (string): Text search across title, description, and tags
- `sortBy` (string): Sort field ("createdAt", "price", "viewsCount", "title")
- `sortOrder` (string): Sort order ("asc", "desc", default: "desc")

Example Request:

```
curl "http://localhost:5000/api/events?category=Music&city=Dubai&limit=5&featured=true"
```

Response:

```
{
  "success": true,
  "message": "Events retrieved successfully",
  "data": {
    "events": [
      {
        "_id": "68a155c40214f9e8ddf64aff",
        "title": "Summer Music Festival",
        "description": "A day of entertainment for the whole family with live music...",
        "category": "Music",
        "type": "Event",
        "venueType": "Outdoor",
        "ageRange": [16, 65],
        "location": {
          "coordinates": {
            "lat": 25.232,
            "lng": 55.2515
          },
          "city": "Dubai",
          "address": "Jumeirah Beach Park"
        },
        "seoMeta": {
```

```

        "title": "Summer Music Festival 2024 - Dubai",
        "description": "Join the biggest music festival in Dubai...",
        "keywords": ["music", "festival", "dubai"]
    },
    "vendorId": {
        "_id": "68a155c30214f9e8ddf64a65",
        "firstName": "Ahmed",
        "lastName": "Music Events",
        "email": "ahmed@musicevent.com"
    },
    "price": 299,
    "currency": "AED",
    "isApproved": true,
    "isFeatured": true,
    "tags": ["music", "festival", "outdoor"],
    "dateSchedule": [
        {
            "date": "2024-07-15T18:00:00.000Z",
            "availableSeats": 10000,
            "_id": "68a155c40214f9e8ddf64b00"
        }
    ],
    "faqs": [
        {
            "question": "What should I bring to the event?",
            "answer": "Please bring a valid ID for entry. Food and drinks available for purchase.",
            "_id": "68a155c40214f9e8ddf64b01"
        }
    ],
    "viewsCount": 2847,
    "images": [
        "https://cloudinary.com/music-festival-banner.jpg",
        "https://cloudinary.com/music-festival-1.jpg"
    ],
    "isDeleted": false,
    "createdAt": "2024-03-15T08:30:00.000Z",
    "updatedAt": "2024-03-16T14:22:15.000Z",
    "affiliateCode": "MUS-FEST-2024"
    }
},
{
    "pagination": {
        "currentPage": 1,
        "totalPages": 3,
        "totalEvents": 15,
        "hasNextPage": true,
        "hasPrevPage": false,
        "limit": 5
    }
}
}

```

GET /api/events/categories

Get all available event categories for filtering.

Authentication: Not required

Example Request:

```
curl http://localhost:5000/api/events/categories
```

Response:

```
{
  "success": true,
  "message": "Event categories retrieved successfully",
  "data": {
    "categories": [
      "Art & Culture",
      "Business",
      "Culture & Heritage",
      "Family & Kids",
      "Food & Dining",
      "Health & Wellness",
      "Music",
      "Sports & Recreation",
      "Technology"
    ]
  }
}
```

GET /api/events/:id

Get detailed information for a single event.

Authentication: Not required

Path Parameters:

- `id` (string): Event MongoDB ObjectId

Example Request:

```
curl http://localhost:5000/api/events/68a155c40214f9e8ddf64aff
```

Response:

```
{
  "success": true,
  "message": "Event retrieved successfully",
  "data": {
    "event": {
      "_id": "68a155c40214f9e8ddf64aff",
      "title": "Summer Music Festival",
      "description": "The biggest music event of the year featuring world-renowned DJs and live bands...",
      "category": "Music",
      "type": "Event",
      "venueType": "Outdoor",
      "ageRange": [16, 65],
      "location": {
        "coordinates": {
          "lat": 25.232,
          "lng": 55.2515
        },
        "city": "Dubai",
        "address": "Dubai Marina Beach, Marina Walk"
      },
      "price": 299,
      "currency": "AED",
      "tags": ["music", "festival", "outdoor", "international"],
      "dateSchedule": [
        {
          "date": "2024-07-15T18:00:00.000Z",
          "availableSeats": 10000,
          "price": 299,
          "_id": "68a155c40214f9e8ddf64b00"
        }
      ],
      "faqs": [
        {
          "question": "What should I bring to the event?",
          "answer": "Please bring a valid ID for entry. Food and drinks will be available for purchase. Chairs and blankets are welcome."
        }
      ]
    }
  }
}
```

```
      "_id": "68a155c40214f9e8ddf64b01"
    },
    {
      "question": "Is parking available?",
      "answer": "Yes, complimentary parking is available at Dubai Marina Mall. Shuttle service
provided.",
      "_id": "68a155c40214f9e8ddf64b02"
    }
  ],
  "images": [
    "https://cloudinary.com/music-festival-banner.jpg",
    "https://cloudinary.com/music-festival-1.jpg",
    "https://cloudinary.com/music-festival-2.jpg"
  ],
  "seoMeta": {
    "title": "Summer Music Festival 2024 - Dubai Marina Beach",
    "description": "Join the biggest music festival in Dubai featuring international artists. Book
your tickets now!",
    "keywords": ["dubai music festival", "marina beach events", "summer concerts"]
  },
  "vendorId": {
    "_id": "68a155c30214f9e8ddf64a65",
    "firstName": "Ahmed",
    "lastName": "Music Events",
    "email": "ahmed@musicevent.com",
    "phone": "+971501234567",
    "avatar": "https://cloudinary.com/vendor-avatar.jpg"
  },
  "viewsCount": 2848,
  "isFeatured": true,
  "isApproved": true,
  "createdAt": "2024-03-15T08:30:00.000Z",
  "updatedAt": "2024-03-16T14:22:15.000Z"
}
}
```

Note: This endpoint automatically increments the `viewsCount` for analytics tracking.

Vendor Event Management

POST /api/events

Create a new event (Vendor authentication required).

Authentication: Required (Vendor role)

Headers:

Authorization: Bearer <access_token> Content-Type: application/json
--

Request Body:

```
{
  "title": "Amazing Concert Night",
  "description": "A wonderful evening of music and entertainment featuring local and international artists.",
  "category": "Music",
  "type": "Event",
  "venueType": "Indoor",
  "ageRange": [18, 65],
  "location": {
    "city": "Dubai",
    "address": "123 Main Street, Downtown Dubai",
    "coordinates": {
      "lat": 25.2048,
      "lng": 55.2708
    }
  },
  "price": 150,
  "currency": "AED",
  "tags": ["music", "concert", "entertainment", "nightlife"],
  "dateSchedule": [{
    "date": "2024-03-15T19:00:00Z",
    "availableSeats": 500,
    "price": 150
  }],
  "faqs": [
    {
      "question": "What time does the event start?",
      "answer": "The event starts at 7:00 PM. Gates open at 6:00 PM."
    }
  ],
  "images": ["image_url_1", "image_url_2"],
  "seoMeta": {
    "title": "Amazing Concert Night 2024 - Dubai",
    "description": "Don't miss the amazing concert night in Dubai",
    "keywords": ["concert", "dubai", "music", "entertainment"]
  }
}
```

Response:

```
{
  "success": true,
  "message": "Event created successfully and is pending admin approval",
  "data": {
    "event": {
      "_id": "new_event_id",
      "title": "Amazing Concert Night",
      "description": "A wonderful evening of music and entertainment...",
      "isApproved": false,
      "status": "PENDING",
      "vendorId": "vendor_id",
      "affiliateCode": "AMZ-CONC-2024",
      "createdAt": "2024-03-16T10:30:00.000Z",
      "updatedAt": "2024-03-16T10:30:00.000Z"
    }
  }
}
```

PUT /api/events/:id

Update an existing event (Event owner authentication required).

Authentication: Required (Event owner only)

Headers:

Authorization: Bearer <access_token>
Content-Type: application/json

Path Parameters:

- `id` (string): Event MongoDB ObjectId

Request Body: Same structure as POST `/api/events` (partial updates allowed)

Response:

```
{
  "success": true,
  "message": "Event updated successfully. Pending admin re-approval due to significant changes.",
  "data": {
    "event": {
      "_id": "event_id",
      "title": "Updated Event Title",
      "isApproved": false,
      "status": "PENDING_UPDATE",
      "updatedAt": "2024-03-16T14:35:00.000Z"
    }
  }
}
```

Notes:

- Major changes (title, description, price, schedule) require admin re-approval
- Minor changes (tags, images) don't require re-approval
- Event remains live during re-approval process unless major pricing/scheduling changes occur

DELETE `/api/events/:id`

Soft delete an event (Event owner authentication required).

Authentication: Required (Event owner only)

Headers:

```
Authorization: Bearer <access_token>
```

Path Parameters:

- `id` (string): Event MongoDB ObjectId

Response:

```
{
  "success": true,
  "message": "Event deleted successfully"
}
```

Note: This performs a soft delete. Event is marked as deleted but remains in database for order history.

GET `/api/events/vendor/my-events`

Get vendor's own events with filtering and pagination.

Authentication: Required (Vendor role)

Headers:

```
Authorization: Bearer <access_token>
```

Query Parameters:

- `page` (number): Page number (default: 1)
- `limit` (number): Items per page (default: 12, max: 50)
- `status` (string): Filter by status ("all", "approved", "pending", "featured", "rejected")
- `sortBy` (string): Sort field ("createdAt", "title", "price", "viewsCount")
- `sortOrder` (string): Sort order ("asc", "desc", default: "desc")
- `search` (string): Search in event titles

Example Request:

```
curl -X GET "http://localhost:5000/api/events/vendor/my-events?status=approved&limit=10" \
-H "Authorization: Bearer YOUR_ACCESS_TOKEN"
```

Response:

```
{
  "success": true,
  "message": "Vendor events retrieved successfully",
  "data": {
    "events": [
      {
        "_id": "event_id",
        "title": "My Amazing Event",
        "category": "Music",
        "price": 150,
        "currency": "AED",
        "isApproved": true,
        "isFeatured": false,
        "viewsCount": 245,
        "dateSchedule": [...],
        "createdAt": "2024-03-15T10:00:00.000Z",
        "updatedAt": "2024-03-16T14:30:00.000Z"
      }
    ],
    "pagination": {
      "currentPage": 1,
      "totalPages": 3,
      "totalEvents": 25,
      "hasNextPage": true,
      "hasPrevPage": false,
      "limit": 10
    }
  }
}
```

GET /api/events/vendor/analytics

Get comprehensive analytics for vendor's events.

Authentication: Required (Vendor role)

Headers:

```
Authorization: Bearer <access_token>
```

Query Parameters:

- `period` (number): Analysis period in days (default: 30, max: 365)
- `eventId` (string): Specific event analytics (optional)

Example Request:

```
curl -X GET "http://localhost:5000/api/events/vendor/analytics?period=90" \
-H "Authorization: Bearer YOUR_ACCESS_TOKEN"
```

Response:

```
{
  "success": true,
  "message": "Vendor analytics retrieved successfully",
  "data": {
    "analytics": {
      "summary": {
        "totalEvents": 25,
        "approvedEvents": 20,
        "pendingEvents": 3,
        "rejectedEvents": 2,
        "featuredEvents": 5,
        "totalViews": 15420,
        "avgViewsPerEvent": 616,
        "totalRevenue": 125000,
        "avgPrice": 245.50
      },
      "performance": {
        "topPerformingEvent": {
          "title": "Summer Music Festival",
          "views": 2847,
          "revenue": 35000
        },
        "categoryBreakdown": {
          "Music": 12,
          "Sports": 8,
          "Technology": 5
        },
        "monthlyTrends": [
          {
            "month": "2024-01",
            "events": 8,
            "views": 3200,
            "revenue": 24000
          }
        ]
      }
    }
  }
}
```

Admin Event Management

GET /api/events/admin/all

Get all events for administrative management.

Authentication: Required (Admin role)

Headers:

```
Authorization: Bearer <access_token>
```

Query Parameters:

- **page (number):** Page number (default: 1)
- **limit (number):** Items per page (default: 12, max: 100)
- **status (string):** Filter by status ("all", "approved", "pending", "featured", "rejected")
- **vendorId (string):** Filter by specific vendor
- **category (string):** Filter by category
- **sortBy (string):** Sort field ("createdAt", "title", "price", "viewsCount")
- **sortOrder (string):** Sort order ("asc", "desc", default: "desc")
- **search (string):** Search in titles, descriptions

Response:


```
{
  "success": true,
  "message": "Admin events retrieved successfully",
  "data": {
    "events": [
      {
        "_id": "event_id",
        "title": "Pending Event Title",
        "vendorId": {
          "_id": "vendor_id",
          "firstName": "John",
          "lastName": "Vendor",
          "email": "john@vendor.com"
        },
        "category": "Music",
        "isApproved": false,
        "isFeatured": false,
        "status": "PENDING",
        "submittedAt": "2024-03-16T10:30:00.000Z"
      }
    ],
    "pagination": {...}
  }
}
```

PUT /api/events/admin/:id/approval

Approve or reject an event submission.

Authentication: Required (Admin role)

Headers:

Authorization: Bearer <access_token>
Content-Type: application/json

Path Parameters:

- `id` (string): Event MongoDB ObjectId

Request Body:

```
{
  "isApproved": true,
  "reason": "Optional rejection reason if disapproving",
  "adminNotes": "Internal admin notes"
}
```

Response (Approval):

```
{
  "success": true,
  "message": "Event approved successfully",
  "data": {
    "event": {
      "_id": "event_id",
      "title": "Event Title",
      "isApproved": true,
      "approvedAt": "2024-03-16T15:30:00.000Z",
      "approvedBy": "admin_id"
    }
  }
}
```

Response (Rejection):

```
{
  "success": true,
  "message": "Event rejected successfully",
  "data": {
    "event": {
      "_id": "event_id",
      "title": "Event Title",
      "isApproved": false,
      "rejectionReason": "Event description needs more detail",
      "rejectedAt": "2024-03-16T15:30:00.000Z",
      "rejectedBy": "admin_id"
    }
  }
}
```

PUT /api/events/admin/:id/featured

Toggle event featured status for enhanced visibility.

Authentication: Required (Admin role)

Headers:

Authorization: Bearer <access_token>
Content-Type: application/json

Path Parameters:

- id (string): Event MongoDB ObjectId

Request Body:

```
{
  "isFeatured": true
}
```

Response:

```
{
  "success": true,
  "message": "Event featured successfully",
  "data": {
    "event": {
      "_id": "event_id",
      "title": "Event Title",
      "isFeatured": true,
      "featuredAt": "2024-03-16T15:45:00.000Z",
      "featuredBy": "admin_id"
    }
  }
}
```

 Event Analytics & Insights

Event View Tracking

- All public event detail views are automatically tracked
- View counts help vendors understand event popularity
- Analytics aggregation runs daily for performance optimization

SEO Optimization

- Events include structured meta tags for search engines
- Automatic generation of SEO-friendly URLs
- Keywords and descriptions optimized for discovery

Performance Monitoring

- Response time tracking for all event endpoints
- Automated scaling based on traffic patterns

- Caching strategies for high-traffic events

Implementation Examples

Frontend Event Listing

```
// Fetch featured events with filtering
const fetchEvents = async (filters = {}) => {
  const params = new URLSearchParams({
    featured: 'true',
    city: 'Dubai',
    limit: 12,
    ...filters
  });

  const response = await fetch(`/api/events?${params}`);
  const data = await response.json();

  return data.success ? data.data.events : [];
};
```

Vendor Event Creation

```
// Create new event with comprehensive data
const createEvent = async (eventData) => {
  const response = await fetch('/api/events', {
    method: 'POST',
    headers: {
      'Authorization': `Bearer ${accessToken}`,
      'Content-Type': 'application/json'
    },
    body: JSON.stringify({
      ...eventData,
      dateSchedule: eventData.dates.map(date => ({
        date: new Date(date).toISOString(),
        availableSeats: eventData.capacity,
        price: eventData.price
      }))
    })
  });

  const result = await response.json();

  if (!result.success) {
    throw new Error(result.message);
  }

  return result.data.event;
};
```

Admin Event Approval Workflow

```
// Bulk approve/reject events
const processEventBatch = async (eventIds, action, reason = '') => {
  const promises = eventIds.map(id =>
    fetch(`/api/events/admin/${id}/approval`, {
      method: 'PUT',
      headers: {
        'Authorization': `Bearer ${adminToken}`,
        'Content-Type': 'application/json'
      },
      body: JSON.stringify({
        isApproved: action === 'approve',
        reason: action === 'reject' ? reason : undefined
      })
    })
  );

  const results = await Promise.allSettled(promises);
  return results.map(result =>
    result.status === 'fulfilled' ? result.value.json() : null
  );
};
```

Related Documentation:

- [Authentication → \(./authentication.md\)](#) - User and role management
- [Orders & Payments → \(orders-payments.md\)](#) - Event booking and payment processing
- [Venues & Vendors → \(venues-vendors.md\)](#) - Venue and vendor management
- [File Uploads → \(file-uploads.md\)](#) - Event image and document uploads
- [Admin APIs → \(admin-apis.md\)](#) - Additional administrative functions